

DTSC: A Dual-Tier Sector-Based Cooperative Mobile Charging Scheme for Wireless Rechargeable Sensor Networks

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Abstract. Energy demand in a convergecast wireless rechargeable sensor network is structurally uneven: nodes near the base station relay for everyone behind them and drain fastest, while distant nodes cost a charger the most to reach. Chargers scheduled independently over the field therefore spend much of their budget on repeated long return trips. We present DTSC, which cuts the field into angular sectors of balanced consumption and assigns each to an inner-outer charger pair. The outer charger never returns to the base station; the inner charger refuels there and transfers energy at a rendezvous point, so the long return leg is paid once per pair rather than once per charger. Across 12,500 simulations DTSC matches three of four baselines on dead nodes at the heaviest load while travelling 46–62% less, a margin holding at every quantile; the fourth travels less still but loses four times as many nodes.

Keywords: mobile charger scheduling, wireless rechargeable sensor networks, cooperative charging, sector partitioning, rendezvous

1 Introduction

Wireless rechargeable sensor networks (WRSNs) keep battery-powered nodes alive by having mobile chargers deliver energy in place rather than by replacing batteries [9, 16]. In a convergecast deployment, where every node forwards its readings toward a base station (BS), the resulting energy demand is not uniform. A node one hop from the BS relays the traffic of its entire subtree and drains far faster than a leaf, so consumption rate varies systematically with position [20, 7]. At the same time the nodes that are cheapest to serve are not the ones that need service most: distant nodes cost the most travel energy to reach.

These two gradients pull in opposite directions, and a scheduler that ignores the structure pays for it twice. If chargers are dispatched independently over the whole field, each must return to the BS whenever its own battery is depleted, and that return leg is paid once per charger per cycle. As the field grows, travel rather than charging comes to dominate the energy budget [11, 12]. Conversely,

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a scheduler that minimises travel by serving only the most urgent requests leaves the rest of the field unattended, and nodes die even though the fleet was busy.

We propose **DTSC** (Dual-Tier Sector-based Cooperative charging), which attacks the two cost sources separately. Sectoring balances *spatial workload*: the field is cut into angular sectors whose total energy consumption rate is approximately equal, so no charger pair inherits a disproportionate share. Dual-tier cooperation attacks *repeated travel*: each sector is served by an inner–outer charger pair in which the outer charger never returns to the BS at all. The inner charger refuels at the BS and meets its partner at a rendezvous point, transferring energy by contact. One return leg per pair thus replaces one per charger, and the saving recurs every cycle.

Our contributions are three. (i) Taking the consumption-rate model as established in prior work [20, 7], we use it as a *partitioning criterion*: angular sectors are cut at equal cumulative consumption rate rather than by geometry or node count, so that sectors carry comparable service load. (ii) We design an inner–outer service partition with relay charging at a rendezvous point, converting the per-charger return cost into a per-pair cost. (iii) We evaluate across 12,500 simulations against three baselines and one ablation within a single framework, reporting travel and reliability together since either metric alone misrepresents the result, and using the ablation to separate the contribution of the two mechanisms.

2 Related Work

Scheduling for one or a few chargers. Early work optimises the visiting order of a single charger and the stay time at each stop [15, 2], later extended with deployment and path optimisation [8, 19] and precedence constraints [10]. Instant-response variants trigger service on demand rather than on a fixed cycle [4], and long-distance charging relaxes the requirement that a charger park beside each node [1]. Temporal and deadline-aware variants predict when a node will die and order service accordingly [11, 12]. These schemes sharpen the ordering problem but leave the return-to-base cost untouched.

Demand-aware and region-based charging. Recognising that consumption is uneven, several works weight service by measured consumption rate [20, 7] or partition the field into regions to localise charger responsibility [23, 13]. Region-based designs reduce interference between chargers, but the partition is usually geometric or population-based rather than load-based, so a region containing many relay-heavy nodes still receives the same share of the fleet as a quiet one.

Cooperative and multi-vehicle charging. Charger-to-charger energy transfer has been explored as a way to extend field time [24, 14], and heterogeneous or multi-type fleets assign different roles to different vehicles [21, 22]. Dynamic and adaptive schedules react to changing demand [5, 17, 18]. In these designs cooperation is opportunistic: which vehicle assists which is decided at run time from the

current state, so every vehicle must remain able to reach the BS and the saving is whatever the encounter pattern happens to yield. The alternative we take up is to fix the roles in advance, which is what permits one vehicle in each pair to be built into the schedule as never reaching the BS at all.

3 Network Model

Sensors are deployed over a disk with the BS at the centre and report over a convergecast tree. For node i we define its energy consumption rate ECR_i as the sum of sensing, transmission, reception and relay costs, where the relay term is proportional to the traffic of the subtree rooted at i . Because subtree size grows toward the BS, ECR_i grows with proximity to the centre; this gradient is the quantity DTSC balances against.

Chargers move at fixed speed and spend movement energy E_{move} per metre from a finite battery. A charger parks at an anchor point and serves every alive node within charging radius r_c , delivering power by the Friis free-space model at conversion efficiency η_{ch} . Contact-based charger-to-charger transfer is available at a rendezvous point at the far higher efficiency $\eta_{mc} = 0.95$, and that gap is what makes the dual-tier arrangement worthwhile: moving energy between two vehicles that already stand next to each other is far cheaper than driving one of them back to the BS for it.

Objective and Problem Statement

Node i is alive at time t while $E_i(t) > E_{min}$, and the network lifetime is the instant the first node fails, $T_{life} = \min_i \{t \mid E_i(t) \leq E_{min}\}$. The definition is deliberately strict: losing one relay node can disconnect its entire subtree from the BS.

Two quantities set the scale of the problem. The aggregate demand is $P_{total} = \sum_i ECR_i$, and the longest cycle the network can tolerate is fixed by its hungriest node,

$$\tau_{buf} = \frac{E_{max} - E_{min}}{\max_i ECR_i}. \quad (1)$$

Any schedule whose cycle exceeds τ_{buf} allows that node to fall below E_{min} before it is served again.

The scheduling problem is then to keep every node above E_{min} , and thereby sustain data collection at the BS, while spending as little movement energy as possible. The decisions this requires are the ones Section 4 makes in turn and Section 5 evaluates: how to balance workload across sectors, where to divide inner from outer service, where pairs should meet, where chargers should stop, and how to keep a pair's two cycles the same length.

4 The DTSC Scheme

The procedure has six stages: fleet dimensioning and pair assignment; ECR-balanced sectoring; inner-outer service partitioning; charging anchor generation;

Algorithm 1 DTSC

Require: node set V with rates ECR_i , BS at the origin, charging radius r_c
once per deployment

- 1: $\tau_{buf} \leftarrow (E_{max} - E_{min}) / \max_i ECR_i$
- 2: $N_{opt} \leftarrow$ least even integer $\geq \max(N_{energy}, N_{time})$; $K \leftarrow N_{opt}/2$
- 3: sort V by polar angle; cut a sector each time the running ECR sum reaches $\frac{1}{K} \sum_i ECR_i$
- 4: assign pair (c_{in}^k, c_{out}^k) to sector S_k , $k = 1 \dots K$
every cycle, each sector independently
- 5: **for** each sector S_k **do**
- 6: $V_{in} \leftarrow$ greedy expansion outward from the BS under the travel, budget and 50% ECR limits
- 7: $V_{out} \leftarrow S_k \setminus V_{in}$
- 8: $A_{in} \leftarrow \text{GREEDYDISKCOVER}(V_{in}, r_c)$; $A_{out} \leftarrow \text{GREEDYDISKCOVER}(V_{out}, r_c)$
- 9: $\pi_{in} \leftarrow \text{TSP}(A_{in})$; $\pi_{out} \leftarrow \text{TSP}(A_{out})$
- 10: $rp \leftarrow$ midpoint of $\arg \min_{a \in A_{in}, b \in A_{out}} \|a - b\|$
- 11: **repeat**
- 12: recompute T_{in}, T_{out} under the current T ; $T \leftarrow \max(T_{in}, T_{out})$
- 13: **until** T changes by less than 5%
- 14: spread each charger's slack over its own anchors in proportion to their ECR share
- 15: c_{in}^k, c_{out}^k traverse π_{in}, π_{out} ; c_{out}^k proceeds to rp
- 16: c_{in}^k meets c_{out}^k at rp , transfers energy by contact, then refuels at the BS
- 17: **end for**

route planning; and cycle-time estimation with relay charging. The first two fix the structure of the deployment and are computed once; the rest are recomputed whenever a pair replans. Algorithm 1 gives the procedure in full; the subsections below explain the three choices in it that are not routine.

4.1 Fleet Dimensioning and Pair Assignment

The fleet size follows from the load rather than being chosen by hand. Taking τ_{buf} from (1) as the deadline the fleet must meet, two lower bounds apply. The first is energetic: over one such interval the network consumes $P_{total}\tau_{buf}$, which the fleet must be able to replace,

$$N_{energy} = \left\lceil \frac{P_{total} \tau_{buf}}{E_m^{max} \eta_{safe}} \right\rceil, \quad (2)$$

where E_m^{max} is one charger's capacity and η_{safe} reserves part of it as margin. The second is temporal: the estimated global travel and charging work must fit inside the same interval,

$$N_{time} = \left\lceil \frac{T_{travel}^{global} + T_{charge}^{global}}{\tau_{buf}} \right\rceil. \quad (3)$$

The fleet is the larger of the two, rounded up to an even number because chargers are deployed in pairs, and the sector count follows from it:

$$N_{opt} = \text{even}(\max(N_{energy}, N_{time})), \quad K = N_{opt}/2. \quad (4)$$

Every baseline in Section 5 is given the same N_{opt} , so no method is compared against another with more vehicles.

4.2 ECR-Balanced Sectoring

The field is divided into K angular sectors, but the cut points are chosen so that the *summed energy consumption rate* of each sector is approximately equal, not its area or its node count. Sorting nodes by angle and sweeping until the cumulative ECR reaches $\frac{1}{K} \sum_i ECR_i$ yields boundaries that give each charger pair comparable work. Since ECR_i rises toward the centre (Section 3), balancing by node count would instead leave every sector with the same number of nodes but hand whichever sector happened to contain more of the relay-heavy inner nodes a systematically larger share of the actual demand.

4.3 Inner–Outer Partition and Relay Charging

Within a sector, nodes are sorted by distance from the BS and admitted to the inner set by greedy expansion under three limits: the travel cost of reaching them, the inner charger’s energy budget, and a ceiling on the fraction of the sector’s total ECR the inner set may hold (50% in our configuration), which keeps enough of the sector’s demand on the outer side to be worth a dedicated vehicle. The remaining nodes form the outer set. Only the inner charger ever visits the BS; the outer charger never returns to it. Fig. 1 illustrates the arrangement.

Each service set is then covered by charging anchors obtained by greedy disk covering, so that every node in the set falls within charging radius of at least one anchor and one stop serves a whole group rather than a single node; each charger solves a TSP over its own anchors to fix a visiting order. The rendezvous point is derived from these anchors rather than fixed in advance: it is the midpoint of the closest inner-anchor/outer-anchor pair, the cheapest place at which the two routes can be made to touch. It is recomputed whenever the pair replans, so it follows the anchors as nodes are served or die.

Each cycle then proceeds through a fixed sequence: both chargers plan and execute their routes, the outer charger returns to the rendezvous point, the inner charger meets it there and transfers energy by contact, and the inner charger returns to the BS to refuel. The pair’s common cycle time is obtained by fixed-point iteration—estimate the cycle, recompute each charger’s travel and charging time under that estimate, take the larger of the two, and repeat until the change falls below 5%. Whichever charger finishes early does not then idle at the rendezvous point: its slack is redistributed as additional stay time across its own anchors in proportion to their ECR share, so the surplus is spent on the hungriest groups rather than on waiting.

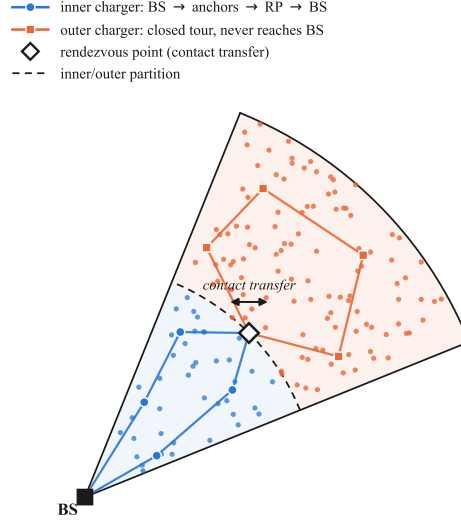


Fig. 1. Inner-outer service partition within one sector, with both routes of the pair. The inner charger’s tour closes at the BS; the outer charger’s closes at the rendezvous point and never reaches the BS, being refuelled there by its partner instead. Anchors are the stops produced by greedy disk covering, not individual nodes.

The design intent is narrow and worth stating plainly: DTSC does not try to minimise travel. It tries to remove one specific repeated cost—the outer charger’s return leg—while keeping service coverage sector-wide. Whether that trade is worthwhile is an empirical question, and Section 5 answers it in both directions.

5 Evaluation

5.1 Setup

The evaluation sweeps five parameters—node count, charger speed, movement energy cost, node battery capacity and charger initial energy—over five values each, for five algorithms, with **100 fixed random seeds** per configuration: 12,500 simulation runs in total, each over a horizon of 10^5 s. For a given parameter value and seed, all five algorithms use the same node distribution and the same total charger count, so the comparison at each plotted point is like-for-like. That count is not a constant: it is derived per network instance by the dimensioning of Section 4.1, so the sector count grows with the load. Across the node-count sweep the derived K runs from 2 at 500 nodes to 17 at 1500—fleets of 4 to 34 chargers—with medians of 3 and 12. Every result below therefore exercises the dimensioning rule over an eightfold span of K rather than at one hand-picked value. When one parameter is swept the others hold their defaults;

Table 1. Simulation parameters. Swept values are given as ranges; the default used when another parameter is being swept is shown in bold.

Field radius	200 m	Node battery E_{max}	20–100 J (100)
BS position	centre	Safe threshold E_{min}	$0.05 E_{max}$
Node count	500–1500 (500)	Charger battery	$0.2\text{--}1.0 \times 10^6$ J (10^6)
Charger speed	2–10 m/s (5)	Charging radius r_c	5 m
Movement cost E_{move}	5–25 J/m (5)	Node charging eff. η_{ch}	0.8
Packet size	4096 bits	Contact transfer eff. η_{mc}	0.95
Sensing rate	2 packets/s	Horizon	10^5 s

the sweeps over speed, movement cost, node capacity and charger energy therefore all run at 500 nodes, and only the node-count sweep reaches 1500. Table 1 lists the values.

The four baselines are re-implementations within the same simulator core, so that differences are attributable to the scheduling policy alone rather than to the surrounding machinery. **Min–Max** follows the spatiotemporal charging scheduler of Hong et al. [6]: AHP node weighting, a spanning tree, graph embedding and clustering, then a TSP route per cluster. **Mean-Shift** is a representative of the reactive, threshold-triggered family [4], grouping outstanding requests by mean-shift clustering [3] and dispatching to the smallest charging delay. **Game-Theory** represents the payoff-driven family in which each charger bids for requests on chargeable energy minus travel cost [5, 17], with ties broken by elimination and a feasibility check before commitment. We reproduce the mechanisms of the latter two rather than a specific published implementation, and report them as family representatives.

Sector-TSP is different in kind and we use it as such: it is an *ablation of DTSC*. It receives the same fleet size and the same ECR-balanced partition, but assigns one charger per sector with no inner–outer split, no rendezvous point and no relay charging, so every charger returns to the BS each cycle. The DTSC/Sector-TSP contrast therefore isolates the dual-tier mechanism from the sectoring that both share. We report total charger travel, network remaining energy, dead node count, and energy usage effectiveness, the fraction of expended charger energy that reaches nodes as charge rather than being spent on movement, $E_{charge}/(E_{charge} + E_{move})$.

5.2 Travel and Reliability Together

Fig. 2 places all five methods on the two axes that matter at the heaviest tested load. Against three of them—Sector-TSP, Mean-Shift and Min–Max—DTSC travels 46–62% less while losing no more nodes. We state that deliberately rather than claiming outright dominance: the dead-node means are 4.16 for DTSC against 4.19, 4.24 and 4.39, and with standard errors of 0.26–0.92 over 100 seeds those gaps are not separable. The defensible claim is equal reliability at roughly half the travel, which is the claim DTSC is designed to support; its travel half

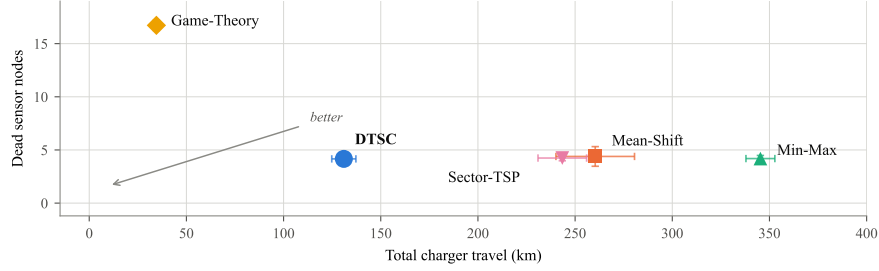


Fig. 2. Travel–reliability trade-off at 1500 nodes; bars are ± 1 standard error over the 100 seeds. DTSC is level with three baselines on dead nodes at roughly half their travel—the vertical intervals overlap, the horizontal ones do not. Only the game-theoretic scheduler travels less, and it loses four times as many nodes.

holds at every quantile of the seed distribution (Section 5.6). The fourth, the game-theoretic scheduler, travels far less than anything else because it serves only selected urgent requests; there the difference is unambiguous—16.7 dead nodes against DTSC’s 4.2, a factor of four and more than twenty standard errors wide.

Reading either axis alone inverts the conclusion. On travel alone the game-theoretic scheduler wins by nearly a factor of four over DTSC, and one would choose it and lose 16.7 nodes; on dead nodes alone the three route-heavy baselines are indistinguishable from DTSC, and one would be free to choose Min–Max and drive 2.6 times as far for it. Only the pair shows what each method actually buys.

5.3 Full Metric Comparison

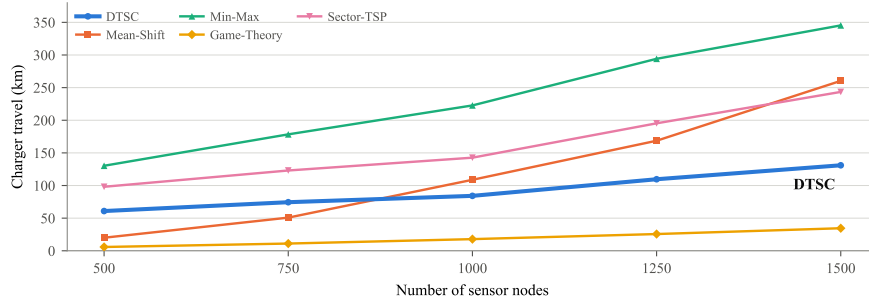
Table 2 gives all four metrics at the heaviest tested load. The pattern that the trade-off plot compresses into two axes is visible here in full. DTSC delivers the second-highest energy usage effectiveness (0.972) behind only the game-theoretic scheduler, whose 0.985 is achieved by simply not travelling—its remaining network energy is the lowest of all five methods at 74.4%, because the nodes it declines to serve are draining. Min–Max retains the most network energy (95.2%) but pays 345 km—2.6 times DTSC’s travel—to do it, and buys no measurable reduction in dead nodes for the difference. The two reliability metrics are not interchangeable: a method can hold the network’s mean energy high and still lose the few nodes whose consumption rate outruns its visit interval, which is why Min–Max leads on remaining energy without leading on dead nodes.

5.4 Which Mechanism Does the Work

DTSC combines two mechanisms, and the Sector-TSP ablation separates them. Both partition the field by cumulative *ECR* and both receive the same fleet; only DTSC pairs its chargers and relays energy between them. At 1500 nodes the pairing alone accounts for a drop from 243.4 to 131.0 km, a 46% reduction,

Table 2. All metrics at 1500 nodes, mean over 100 seeds. Travel lower is better; dead nodes lower is better; remaining energy and effectiveness higher is better.

Method	Travel (km)	Dead nodes	Remaining energy (%)	Effectiveness
DTSC	131.0	4.16	93.0	0.972
Mean-Shift	260.4	4.39	81.1	0.904
Min-Max	345.3	4.19	95.2	0.872
Game-Theory	34.6	16.72	74.4	0.985
Sector-TSP	243.4	4.24	93.9	0.946

**Fig. 3.** Charger travel versus network size, 100-seed means. DTSC scales sub-linearly ($2.2\times$ from 500 to 1500 nodes) where Mean-Shift grows $13.0\times$.

at 4.16 dead nodes against 4.24—that is, at no measurable reliability cost. It also lifts effectiveness from 0.946 to 0.972, because the energy no longer spent on returning is energy available to deliver. The one column that moves the other way is remaining network energy, 93.0% against 93.9%: a single charger covering a whole narrow sector tops nodes up slightly more evenly than a pair covering a wider one. The same pattern holds at 500 nodes, 60.9 against 98.1 km.

Sectoring alone is therefore not what produces the headline result. Sector-TSP has the identical ECR-balanced partition and still travels 86% further than DTSC; the saving comes from removing the outer charger’s return leg, which is the claim Section 5 set out to test.

5.5 Scaling and Sensitivity

Fig. 3 explains where the advantage originates. From 500 to 1500 nodes DTSC’s travel grows by a factor of 2.2, while Mean-Shift’s grows by 13.0. Mean-Shift is the cheapest of all reactive methods at 500 nodes and the second most expensive at 1500; its per-charger return trips multiply with the number of charging events. DTSC starts higher, because a two-tier pair must also position itself for rendezvous, but that overhead is paid once per cycle regardless of scale.

Movement cost is a separate axis and needs care in reading. Across the tested range of E_{move} (5–25 J/m, at 500 nodes) every method’s travel is flat—DTSC at

Table 3. Charger travel over 100 seeds at 1500 nodes, as median, 90th percentile and worst seed. DTSC is below the three route-heavy baselines at every quantile, and Mean-Shift’s mean is set by a small number of divergent runs rather than by its typical behaviour.

Method	Median (km)	p90 (km)	Worst seed (km)
DTSC	114	240	352
Mean-Shift	225	350	1887
Min–Max	340	460	520
Game-Theory	35	42	45
Sector-TSP	216	459	590

roughly 61 km, Mean-Shift at 20 km, Sector-TSP at 98 km—because none of the five schedulers feeds the price of movement back into its routing decisions. Flat travel is thus a property of the comparison and not evidence for DTSC. What does move is effectiveness: as E_{move} rises from 5 to 25 J/m, DTSC’s falls from 0.930 to 0.728 while Min–Max’s falls from 0.815 to 0.468 and Sector-TSP’s from 0.887 to 0.614, so DTSC’s margin over the route-heavy baselines widens from 0.04–0.12 to 0.11–0.26. A shorter route is cheaper to inflate.

5.6 Variability Across Seeds

A mean can be set by a few extreme runs, and with 100 seeds per point we can check whether it is. Table 3 gives the median, the 90th percentile and the worst seed for charger travel at 1500 nodes. The result survives the change of statistic without qualification: DTSC is below all three route-heavy baselines at every one of the three quantiles, so the 46–62% figure is not an artifact of averaging.

The table also shows which method is the least predictable of the five. Mean-Shift’s median is 225 km but its worst seed is 1887 km, more than eight times that and five times DTSC’s worst: the reactive schedule diverges outright on a small number of layouts rather than degrading smoothly across them. DTSC’s own spread is wide in relative terms, 114 km to 352 km, but it is bounded, because each cycle is laid out in advance by the sector partition instead of being driven by the order in which requests happen to arrive.

The scaling result of Section 5.5 does not rest on these tails either. Taking medians in place of means, DTSC’s travel still grows $1.8\times$ from 500 to 1500 nodes against Mean-Shift’s $11.6\times$.

5.7 When DTSC Is Not the Right Choice

The mechanism buys reliability with a fixed structural overhead, so it earns its keep only when that overhead is small relative to the travel it removes. At 500 nodes DTSC still has the better dead-node count—0.01 against Mean-Shift’s 1.12—but the absolute stakes are small, roughly one node in five hundred, and DTSC pays three times Mean-Shift’s travel for it. Whether that is worth buying

depends on how expensive a single loss is, and in many deployments it is not. The reservation is specific to the two low-travel methods: against Sector-TSP and Min-Max, DTSC is ahead on both axes at 500 nodes as well, 0.01 dead nodes at 61 km versus 0.09 at 98 km and 0.16 at 130 km. The case for the mechanism strengthens as the network grows, which is the regime in which the repeated return leg comes to dominate.

6 Conclusion

DTSC balances sectors by consumption rate rather than by geometry, and pairs an inner and an outer charger so that the outer vehicle is freed from the base station entirely. Across 12,500 simulations it matches three of four baselines on dead-node count at the heaviest load while travelling 46–62% less, and its travel scales sub-linearly where a reactive baseline’s grows thirteenfold. The travel margin holds at every quantile of the seed distribution and not only at the mean: over 100 seeds DTSC’s worst run drives 352 km, where the reactive baseline’s drives 1887 km. The game-theoretic baseline travels less still, but loses four times as many nodes; which of the two is preferable depends on whether node loss or charger energy is the binding constraint.

Two limitations bound these results. First, the evaluation is simulation-only, so localisation error and terrain are absent, and contact-based transfer is modelled at a fixed efficiency rather than measured. Second, the sector count is derived once per deployment and then held. The sweep exercises that rule over $K = 2$ to 17, which shows it yields workable fleets across the span but not that the value it selects is the best available; establishing that, and adapting K online as the consumption gradient shifts, is the natural next step.

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